

Name: \_\_\_\_\_



# ***New York State Testing Program***

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## **Mathematics Test Session 1**

**Grade 5**

**Spring 2025**

|                           |
|---------------------------|
| <b>RELEASED QUESTIONS</b> |
|---------------------------|

# Grade 5 Mathematics Reference Sheet

## CONVERSIONS

1 yard = 3 feet  
1 mile = 5,280 feet  
1 mile = 1,760 yards

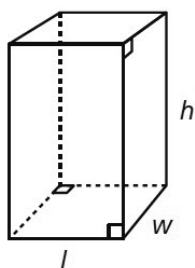
1 cup = 8 fluid ounces  
1 pint = 2 cups  
1 quart = 2 pints  
1 gallon = 4 quarts  
1 liter = 1,000 milliliters

1 pound = 16 ounces  
1 ton = 2,000 pounds  
1 kilogram = 1,000 grams

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## FORMULAS AND FIGURES

### Right Rectangular Prism



$$V = l \times w \times h$$
$$V = B \times h$$

# Session 1



## TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

- Read each question carefully. Take your time.
- You have a ruler, a protractor, and a reference sheet that you can use on the test if they help you answer the question.

**3**

What is the quotient of  $2,550 \div 25$  ?

**A** 100

**B** 102

**C** 105

**D** 120

**5**

Which number has a 2 in the tens place?

**A** 0.26

**B** 2.09

**C** 3.726

**D** 425.9

**GO ON**

**9** Which expression is equivalent to  $\frac{3}{4} \times 7$ ?

**A**  $3 \times 4 \div 7$

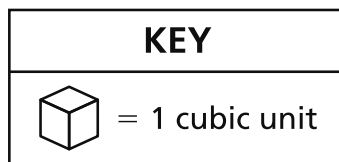
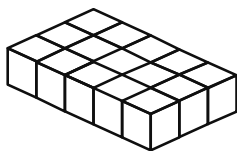
**B**  $3 \times 7 \div 4$

**C**  $3 \div 4 \div 7$

**D**  $3 \times 7 \times 4$

10

The first layer of a right rectangular prism is shown below. Each small cube has a volume of 1 cubic unit.



The height of the entire right rectangular prism is 6 unit cubes. What is the volume, in cubic units, of this prism?

- A 15
- B 23
- C 60
- D 90

11

A baker has  $\frac{1}{4}$  of a box of muffin mix. He pours all the muffin mix equally into 3 bowls.

What fraction of the whole box of muffin mix is in each bowl?

- A  $\frac{1}{12}$
- B  $\frac{3}{4}$
- C  $2\frac{3}{4}$
- D  $3\frac{1}{4}$

**GO ON**

- 13** Jamie has aquarium rocks that come in bags that weigh  $2\frac{2}{5}$  pounds each. She has  $1\frac{1}{2}$  bags of rocks. What is the total weight, in pounds, of the aquarium rocks that Jamie has?

- A**  $1\frac{3}{5}$
- B**  $3\frac{3}{7}$
- C**  $3\frac{3}{5}$
- D**  $3\frac{9}{10}$

- 14** What is the value of  $\frac{34}{100} + \frac{2}{10}$  ?

- A**  $\frac{54}{100}$
- B**  $\frac{54}{10}$
- C**  $\frac{36}{100}$
- D**  $\frac{36}{10}$

**15**

Which shape always has four sides of equal length?

- A** rectangle
- B** rhombus
- C** parallelogram
- D** trapezoid

***GO ON***



**20**

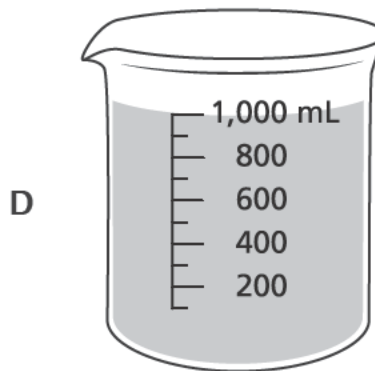
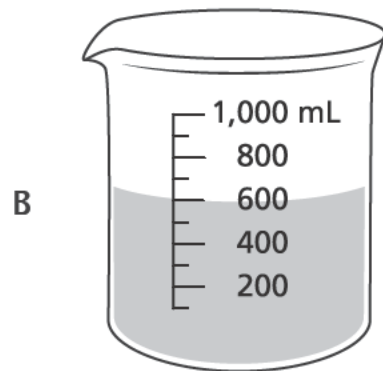
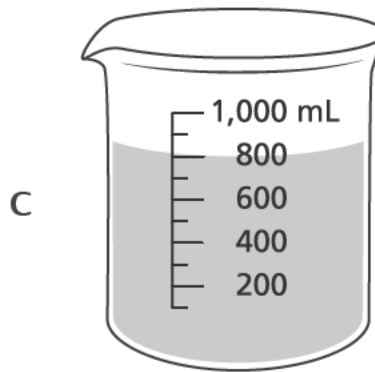
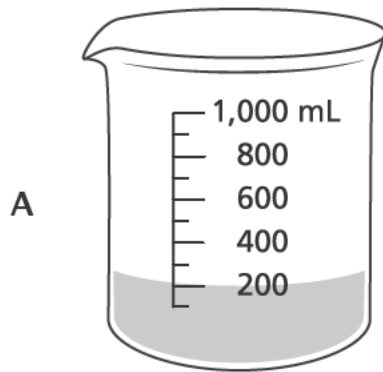
What is 63.4368 rounded to the nearest hundredth?

- A 63.4
- B 63.43
- C 63.44
- D 63.437

***GO ON***

24

A container had 1 liter of water. Exactly 800 milliliters of the water were removed from the container. Which picture shows the amount of water remaining in the container?



25

Lori uses 12 pounds of turkey to make 60 sandwiches. Each sandwich has the same amount of turkey. What is the total amount of turkey in each sandwich?

A  $\frac{1}{6}$  pound

B  $\frac{1}{5}$  pound

C 5 pounds

D 6 pounds

**GO ON**

- 26** Sheri walks  $1\frac{1}{3}$  miles to the store. From the store she walks  $\frac{2}{5}$  of a mile to a friend's house.

What is the total distance, in miles, that Sheri walks?

- A**  $\frac{8}{15}$
- B**  $\frac{6}{8}$
- C**  $1\frac{3}{8}$
- D**  $1\frac{11}{15}$

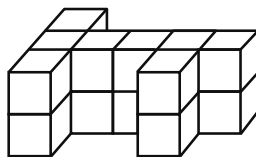
28

In which number does the digit 6 represent a value that is one tenth the value represented by the digit 6 in the number 506.42 ?

- A 504.26
- B 540.62
- C 560.42
- D 604.25

29

The figure shown below is made of unit cubes. The bottom layer of the figure is identical to the top layer.



What is the volume, in cubic units, of the figure?

- A 16
- B 20
- C 24
- D 30

**GO ON**

# Session 2



## TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

- Read each question carefully. Take your time.
- You have a ruler, a protractor, and a reference sheet that you can use on the test if they help you answer the question.
- Be sure to show your work when asked.
- Be sure to explain your answer when asked.

**31**

A company owner spends \$1,488 for tickets to a baseball game. The price of each ticket is \$24. How many tickets does the company owner buy?

- A 62
- B 68
- C 74
- D 75

**32**

An art teacher makes green paint by combining 2 quarts of yellow paint and 3 pints of blue paint. How much green paint, in cups, does the art teacher make?

- A 7
- B 10
- C 14
- D 20

**33**

Which expression is equivalent to  $1\frac{5}{14} - \frac{3}{4}$ ?

- A  $\frac{15}{14} - \frac{13}{14}$
- B  $\frac{33}{28} - \frac{3}{28}$
- C  $\frac{38}{28} - \frac{21}{28}$
- D  $\frac{19}{56} - \frac{3}{56}$

**GO ON**

**34**

Jaylah is buying her favorite candy at a store. The cost for each piece of candy is \$0.63. Jaylah buys 5 pieces of candy. She pays with a \$5.00 bill. What is the total amount of money she should get in change?

- A \$1.55
- B \$1.85
- C \$3.05
- D \$3.15

**35**

Which statement about parallelograms is true?

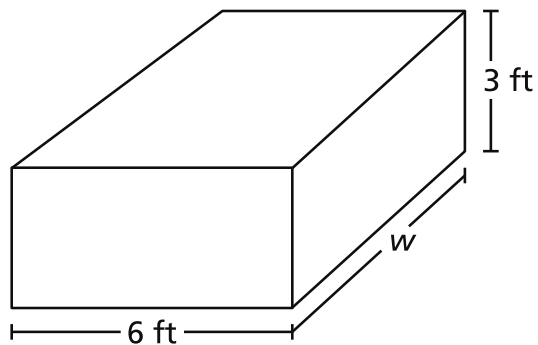
- A All parallelograms are squares.
- B All parallelograms are rectangles.
- C All parallelograms are rhombuses.
- D All parallelograms are quadrilaterals.

**GO ON**

36

This question is worth 1 credit.

A right rectangular prism is shown below.



The volume of the prism is 90 cubic feet. What is the width,  $w$ , in feet, of the prism?

**Answer**  $w =$  \_\_\_\_\_ feet

**GO ON**



37

This question is worth 1 credit.

Last week, Nancy hiked  $7\frac{3}{4}$  miles. This week, she swam  $\frac{2}{3}$  the distance she hiked last week.

How many miles did Nancy swim this week?

Answer \_\_\_\_\_ miles

**GO ON**

**38**

**This question is worth 1 credit.**

Leia has 5 pounds of chocolate that she will put into bags. She puts  $\frac{1}{3}$  pound of the chocolate into each bag. Into how many bags does Leia put the chocolate?

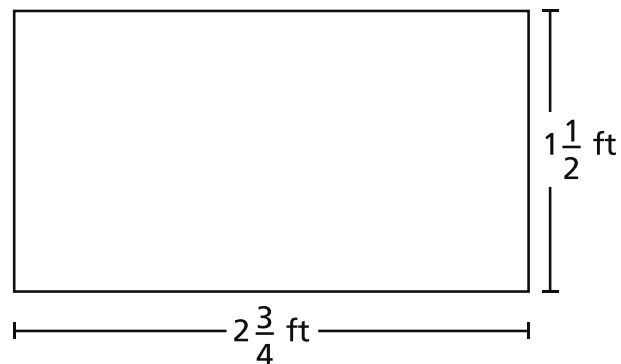
**Answer** \_\_\_\_\_ bags

**GO ON**

39

This question is worth 2 credits.

A diagram of a rectangle with given dimensions is shown below.



What is the area, in square feet, of the rectangle?

*Show your work.*

*Answer* \_\_\_\_\_ square feet

**GO ON**

40

This question is worth 2 credits.

Write a comparison statement using  $>$ ,  $<$ , or  $=$  that shows the relationship between the numbers 157.890 and 157.809.

*Explain how you know your answer is correct.*

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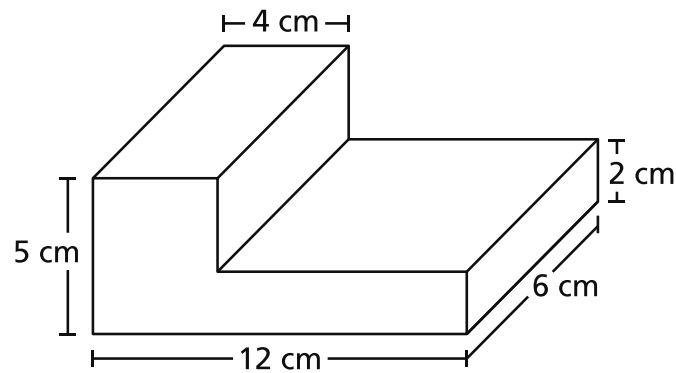
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**GO ON**

41

This question is worth 2 credits.

Two right rectangular prisms were combined to make the figure shown below.



What is the total volume, in cubic centimeters, of the figure?

*Show your work.*

**Answer** \_\_\_\_\_ cubic centimeters

**GO ON**

**42**

**This question is worth 2 credits.**

A teacher writes the equation  $6 \times \frac{3}{3} = 6$  on the board. A student says the equation is wrong because multiplying 6 by a fraction results in a product that is less than 6. Is the student correct?

***Explain your answer.***

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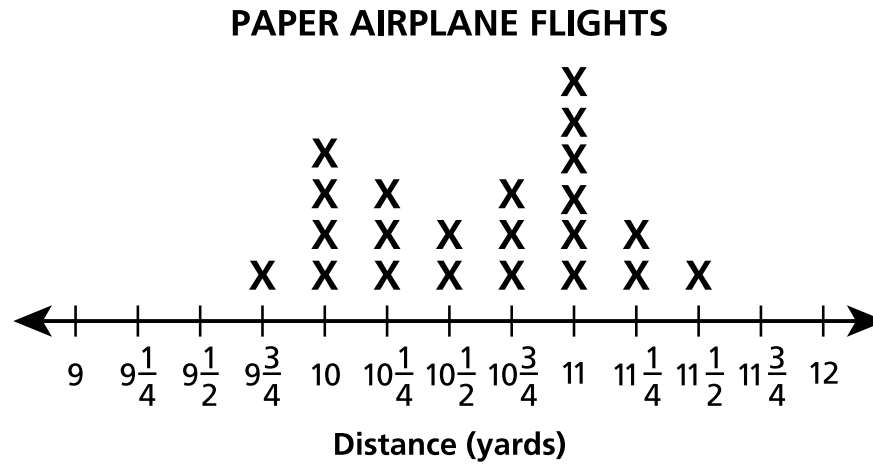
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***GO ON***

43

This question is worth 2 credits.

The Science Club tests paper airplane designs by measuring how far they fly. The results are recorded in the line plot shown below.



What is the difference, in yards, between the longest and shortest flights?

*Show your work.*

Answer \_\_\_\_\_ yards

**GO ON**

44

**This question is worth 3 credits.**

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

***Show your work.***

**Answer** \_\_\_\_\_ miles

**STOP**



**THE STATE EDUCATION DEPARTMENT**  
**THE UNIVERSITY OF THE STATE OF NEW YORK / ALBANY, NY 12234**  
**2025 Mathematics Tests Map to the Standards**  
**Grade 5**

| Question  | Type                 | Key | Points | Standard                      | Cluster                           | Subscore                          | Secondary Standard(s)       | Multiple Choice Questions                               | Constructed-Response Questions |                                                         |
|-----------|----------------------|-----|--------|-------------------------------|-----------------------------------|-----------------------------------|-----------------------------|---------------------------------------------------------|--------------------------------|---------------------------------------------------------|
|           |                      |     |        |                               |                                   |                                   |                             | Percentage of Students Who Answered Correctly (P-Value) | Average Points Earned          | P-Value (Average Points Earned ÷ Total Possible Points) |
| Session 1 |                      |     |        |                               |                                   |                                   |                             |                                                         |                                |                                                         |
| 3         | Multiple Choice      | B   | 1      | NGLS.Math.Content.NY-5.NBT.6  | Number and Operations in Base Ten | Number and Operations in Base Ten |                             | 0.72                                                    |                                |                                                         |
| 5         | Multiple Choice      | D   | 1      | NGLS.Math.Content.NY-5.NBT.1  | Number and Operations in Base Ten | Number and Operations in Base Ten |                             | 0.62                                                    |                                |                                                         |
| 9         | Multiple Choice      | B   | 1      | NGLS.Math.Content.NY-5.NF.4a  | Number and Operations - Fractions | Number and Operations - Fractions |                             | 0.53                                                    |                                |                                                         |
| 10        | Multiple Choice      | D   | 1      | NGLS.Math.Content.NY-5.MD.5a  | Measurement and Data              | Measurement and Data              |                             | 0.49                                                    |                                |                                                         |
| 11        | Multiple Choice      | A   | 1      | NGLS.Math.Content.NY-5.NF.7c  | Number and Operations - Fractions | Number and Operations - Fractions |                             | 0.67                                                    |                                |                                                         |
| 13        | Multiple Choice      | C   | 1      | NGLS.Math.Content.NY-5.NF.6   | Number and Operations - Fractions | Number and Operations - Fractions |                             | 0.28                                                    |                                |                                                         |
| 14        | Multiple Choice      | A   | 1      | NGLS.Math.Content.NY-4.NF.5   | Number and Operations - Fractions | Number and Operations - Fractions |                             | 0.62                                                    |                                |                                                         |
| 15        | Multiple Choice      | B   | 1      | NGLS.Math.Content.NY-5.G.4    | Geometry                          |                                   |                             | 0.47                                                    |                                |                                                         |
| 20        | Multiple Choice      | C   | 1      | NGLS.Math.Content.NY-5.NBT.4  | Number and Operations in Base Ten | Number and Operations in Base Ten |                             | 0.57                                                    |                                |                                                         |
| 24        | Multiple Choice      | A   | 1      | NGLS.Math.Content.NY-4.MD.2b  | Measurement and Data              | Measurement and Data              | NGLS.Math.Content.NY-4.MD.1 | 0.75                                                    |                                |                                                         |
| 25        | Multiple Choice      | B   | 1      | NGLS.Math.Content.NY-5.NF.3   | Number and Operations - Fractions | Number and Operations - Fractions |                             | 0.35                                                    |                                |                                                         |
| 26        | Multiple Choice      | D   | 1      | NGLS.Math.Content.NY-5.NF.2   | Number and Operations - Fractions | Number and Operations - Fractions |                             | 0.70                                                    |                                |                                                         |
| 28        | Multiple Choice      | B   | 1      | NGLS.Math.Content.NY-5.NBT.1  | Number and Operations in Base Ten | Number and Operations in Base Ten |                             | 0.61                                                    |                                |                                                         |
| 29        | Multiple Choice      | A   | 1      | NGLS.Math.Content.NY-5.MD.4   | Measurement and Data              | Measurement and Data              |                             | 0.72                                                    |                                |                                                         |
| Session 2 |                      |     |        |                               |                                   |                                   |                             |                                                         |                                |                                                         |
| 31        | Multiple Choice      | A   | 1      | NGLS.Math.Content.NY-5.NBT.6  | Number and Operations in Base Ten | Number and Operations in Base Ten |                             | 0.79                                                    |                                |                                                         |
| 32        | Multiple Choice      | C   | 1      | NGLS.Math.Content.NY-5.MD.1   | Measurement and Data              | Measurement and Data              |                             | 0.50                                                    |                                |                                                         |
| 33        | Multiple Choice      | C   | 1      | NGLS.Math.Content.NY-5.NF.1   | Number and Operations - Fractions | Number and Operations - Fractions |                             | 0.58                                                    |                                |                                                         |
| 34        | Multiple Choice      | B   | 1      | NGLS.Math.Content.NY-5.NBT.7  | Number and Operations in Base Ten | Number and Operations in Base Ten |                             | 0.66                                                    |                                |                                                         |
| 35        | Multiple Choice      | D   | 1      | NGLS.Math.Content.NY-5.G.3    | Geometry                          |                                   |                             | 0.67                                                    |                                |                                                         |
| 36        | Constructed Response | n/a | 1      | NGLS.Math.Content.NY-5.MD.5b  | Measurement and Data              | Measurement and Data              |                             |                                                         | 0.57                           | 0.57                                                    |
| 37        | Constructed Response | n/a | 1      | NGLS.Math.Content.NY-5.NF.6   | Number and Operations - Fractions | Number and Operations - Fractions |                             |                                                         | 0.18                           | 0.18                                                    |
| 38        | Constructed Response | n/a | 1      | NGLS.Math.Content.NY-5.NF.7c  | Number and Operations - Fractions | Number and Operations - Fractions |                             |                                                         | 0.54                           | 0.54                                                    |
| 39        | Constructed Response | n/a | 2      | NGLS.Math.Content.NY-5.NF.4b  | Number and Operations - Fractions | Number and Operations - Fractions |                             |                                                         | 0.86                           | 0.43                                                    |
| 40        | Constructed Response | n/a | 2      | NGLS.Math.Content.NY-5.NBT.3b | Number and Operations in Base Ten | Number and Operations in Base Ten |                             |                                                         | 0.94                           | 0.47                                                    |
| 41        | Constructed Response | n/a | 2      | NGLS.Math.Content.NY-5.MD.5c  | Measurement and Data              | Measurement and Data              |                             |                                                         | 0.76                           | 0.38                                                    |
| 42        | Constructed Response | n/a | 2      | NGLS.Math.Content.NY-5.NF.5b  | Number and Operations - Fractions | Number and Operations - Fractions |                             |                                                         | 1.24                           | 0.62                                                    |
| 43        | Constructed Response | n/a | 2      | NGLS.Math.Content.NY-5.MD.2   | Measurement and Data              | Measurement and Data              |                             |                                                         | 0.72                           | 0.36                                                    |
| 44        | Constructed Response | n/a | 3      | NGLS.Math.Content.NY-5.NBT.7  | Number and Operations in Base Ten | Number and Operations in Base Ten |                             |                                                         | 1.17                           | 0.39                                                    |

\*This item map is intended to identify the primary analytic skills necessary to successfully answer each question. However, some questions measure proficiencies described in multiple standards, including a balanced combination of procedural and conceptual understanding.

### 1-Credit Constructed-Response Rubric

|                   |                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1 Credit</b>   | A 1-credit response is a <b>correct answer</b> to the question which indicates a thorough understanding of mathematical concepts and/or procedures. |
| <b>0 Credits*</b> | A 0-credit response is incorrect, irrelevant, or incoherent.                                                                                        |

\* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

### 2-Credit Constructed-Response Holistic Rubric

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2 Credits</b>  | <p>A 2-credit response includes the correct solution to the question and demonstrates a thorough understanding of the mathematical concepts and/or procedures in the task.</p> <p>This response</p> <ul style="list-style-type: none"><li>• indicates that the student has completed the task correctly, using mathematically sound procedures</li><li>• contains sufficient work to demonstrate a thorough understanding of the mathematical concepts and/or procedures</li><li>• may contain inconsequential errors that do not detract from the correct solution and the demonstration of a thorough understanding</li></ul> |
| <b>1 Credit</b>   | <p>A 1-credit response demonstrates only a partial understanding of the mathematical concepts and/or procedures in the task.</p> <p>This response</p> <ul style="list-style-type: none"><li>• correctly addresses only some elements of the task</li><li>• may contain an incorrect solution but applies a mathematically appropriate process</li><li>• may contain the correct solution but required work is incomplete</li></ul>                                                                                                                                                                                              |
| <b>0 Credits*</b> | A 0-credit response is incorrect, irrelevant, incoherent, or contains a correct solution obtained using an obviously incorrect procedure. Although some elements may contain correct mathematical procedures, holistically they are not sufficient to demonstrate even a limited understanding of the mathematical concepts embodied in the task.                                                                                                                                                                                                                                                                               |

\* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

### 3-Credit Constructed-Response Holistic Rubric

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3 Credits</b>  | <p>A 3-credit response includes the correct solution(s) to the question and demonstrates a thorough understanding of the mathematical concepts and/or procedures in the task.</p> <p>This response</p> <ul style="list-style-type: none"> <li>• indicates that the student has completed the task correctly, using mathematically sound procedures</li> <li>• contains sufficient work to demonstrate a thorough understanding of the mathematical concepts and/or procedures</li> <li>• may contain inconsequential errors that do not detract from the correct solution(s) and the demonstration of a thorough understanding</li> </ul>                                                          |
| <b>2 Credits</b>  | <p>A 2-credit response demonstrates a partial understanding of the mathematical concepts and/or procedures in the task.</p> <p>This response</p> <ul style="list-style-type: none"> <li>• appropriately addresses most but not all aspects of the task using mathematically sound procedures</li> <li>• may contain an incorrect solution but provides sound procedures, reasoning, and/or explanations</li> <li>• may reflect some minor misunderstanding of the underlying mathematical concepts and/or procedures</li> </ul>                                                                                                                                                                    |
| <b>1 Credit</b>   | <p>A 1-credit response demonstrates only a limited understanding of the mathematical concepts and/or procedures in the task.</p> <p>This response</p> <ul style="list-style-type: none"> <li>• may address some elements of the task correctly but reaches an inadequate solution and/or provides reasoning that is faulty or incomplete</li> <li>• exhibits multiple flaws related to misunderstanding of important aspects of the task, misuse of mathematical procedures, or faulty mathematical reasoning</li> <li>• reflects a lack of essential understanding of the underlying mathematical concepts</li> <li>• may contain the correct solution(s) but required work is limited</li> </ul> |
| <b>0 Credits*</b> | <p>A 0-credit response is incorrect, irrelevant, incoherent, or contains a correct solution obtained using an obviously incorrect procedure. Although some elements may contain correct mathematical procedures, holistically they are not sufficient to demonstrate even a limited understanding of the mathematical concepts embodied in the task.</p>                                                                                                                                                                                                                                                                                                                                           |

\* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

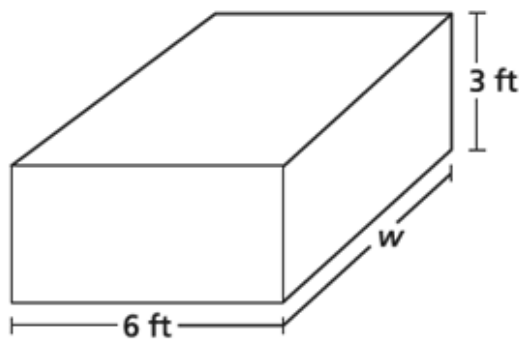
### **1-Credit Constructed-Response Mathematics Scoring Policies**

1. The student is **not** required to show work for a 1-credit constructed-response question, therefore, any work shown will **not** be scored. A clearly identified correct response should still receive full credit.
2. If the student clearly identifies a correct answer but fails to write that answer in the answer space, the student should still receive full credit.
3. If the student provides one legible response (and one response only), the rater should score the response, even if it has been crossed out.
4. If the student has written more than one response but has crossed some out, the rater should score only the response that has **not** been crossed out.
5. If the student provides more than one response but does not indicate which response is to be considered the correct response and none have been crossed out, the student shall not receive credit.
6. If the student does not provide the answer in the form as directed in the question, the student will not receive credit.
7. In questions requiring number sentences, the number sentences must be written horizontally.
8. When measuring angles with a protractor, there is a  $\pm 5$  degrees deviation allowed of the true measure.
9. Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted). This is not to be confused with a score of zero wherein the student does respond to part or all of the question, but that work results in a score of zero.

## 2- and 3-Credit Constructed-Response Mathematics Scoring Policies

1. If a student shows the work in other than a designated “Show your work” or “Explain” area, that work should still be scored.
2. If the question requires students to show their work, and the student shows appropriate work and clearly identifies a correct answer but fails to write that answer in the answer space, the student should still receive full credit.
3. If students are directed to show work or provide an explanation, a correct answer with **no** work shown or **no** explanation provided, receives **no** credit.
4. If students are **not** directed to show work, any work shown will **not** be scored. This applies to questions that do **not** ask for any work and questions that ask for work for one part and do **not** ask for work in another part.
5. If the student provides one legible response (and one response only), the rater should score the response, even if it has been crossed out.
6. If the student has written more than one response but has crossed some out, the rater should score only the response that has **not** been crossed out.
7. If the student provides more than one response, but does not indicate which response is to be considered the correct response and none have been crossed out, the student shall not receive full credit.
8. Trial-and-error responses are **not** subject to Scoring Policy #6 above, since crossing out is part of the trial-and-error process.
9. If a response shows repeated occurrences of the same conceptual error within a question, the conceptual error should **not** be considered more than once in gauging the demonstrated level of understanding.
10. In questions requiring number sentences, the number sentences must be written horizontally.
11. When measuring angles with a protractor, there is a +/- 5 degrees deviation allowed of the true measure.
12. Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted). This is not to be confused with a score of zero wherein the student does respond to part or all of the question but that work results in a score of zero.

A right rectangular prism is shown below.



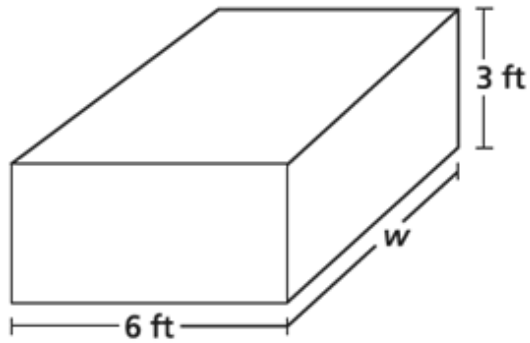
The volume of the prism is 90 cubic feet. What is the width,  $w$ , in feet, of the prism?

**Answer**  $w =$  \_\_\_\_\_ feet

## EXEMPLARY RESPONSE

36

A right rectangular prism is shown below.



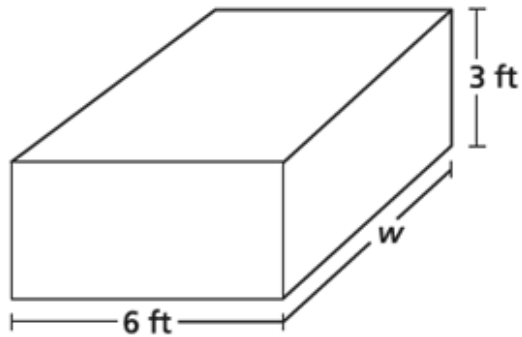
The volume of the prism is 90 cubic feet. What is the width,  $w$ , in feet, of the prism?

**Answer**  $w =$  5 feet

# GUIDE PAPER 1

36

A right rectangular prism is shown below.



The volume of the prism is 90 cubic feet. What is the width,  $w$ , in feet, of the prism?

$$\begin{aligned} V &= 6 \times W \times 3 \\ 6 \times 3 &= 18 \text{ ft} \\ 90 \div 18 &= 5 \text{ ft} \\ \text{Answer } w &= W = 5 \text{ ft} \end{aligned}$$
 feet

**Score Credit 1 (out of 1 credit)**

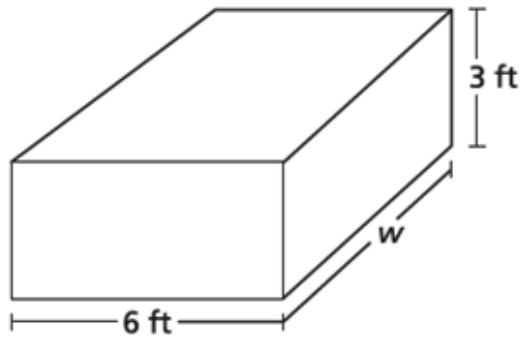
A correct answer is provided.



## GUIDE PAPER 2

36

A right rectangular prism is shown below.



The volume of the prism is 90 cubic feet. What is the width,  $w$ , in feet, of the prism?

Answer  $w =$   feet

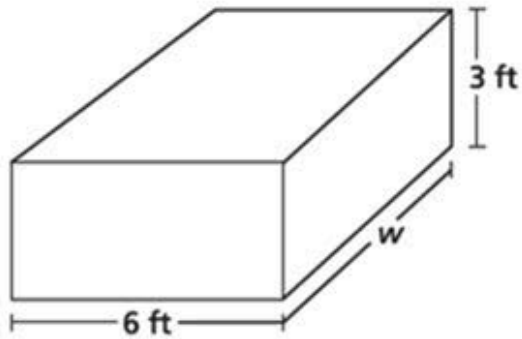
**Score Credit 1 (out of 1 credit)**

A correct answer is provided.

## GUIDE PAPER 3

36

A right rectangular prism is shown below.



The volume of the prism is 90 cubic feet. What is the width,  $w$ , in feet, of the prism?

Answer  $w = 6 \times 3 = 18 \times 5 = 90$  feet

**Score Credit 0 (out of 1 credit)**

The answer is not clearly identified.

Last week, Nancy hiked  $7\frac{3}{4}$  miles. This week, she swam  $\frac{2}{3}$  the distance she hiked last week.

How many miles did Nancy swim this week?

*Answer* \_\_\_\_\_ miles

## EXEMPLARY RESPONSE

37

Last week, Nancy hiked  $7\frac{3}{4}$  miles. This week, she swam  $\frac{2}{3}$  the distance she hiked last week.

How many miles did Nancy swim this week?

Answer  $5\frac{1}{6}$  OR  $\frac{31}{6}$  miles  
*OR Equivalent*

# GUIDE PAPER 1

37

Last week, Nancy hiked  $7\frac{3}{4}$  miles. This week, she swam  $\frac{2}{3}$  the distance she hiked last week.

How many miles did Nancy swim this week?

*Answer*  $7\frac{3}{4} \times \frac{2}{3} = 5\frac{2}{12}$  miles

**Score Credit 1 (out of 1 credit)**

A correct answer is provided.

## GUIDE PAPER 2

37

Last week, Nancy hiked  $7\frac{3}{4}$  miles. This week, she swam  $\frac{2}{3}$  the distance she hiked last week.

How many miles did Nancy swim this week?

*Answer*

$$5\frac{1}{6}$$

miles

**Score Credit 1 (out of 1 credit)**

A correct answer is provided.

## GUIDE PAPER 3

37

Last week, Nancy hiked  $7\frac{3}{4}$  miles. This week, she swam  $\frac{2}{3}$  the distance she hiked last week.

How many miles did Nancy swim this week?

*Answer*

$\frac{34}{6}$

miles

**Score Credit 0 (out of 1 credit)**

An incorrect answer is provided.

Leia has 5 pounds of chocolate that she will put into bags. She puts  $\frac{1}{3}$  pound of the chocolate into each bag. Into how many bags does Leia put the chocolate?

*Answer* \_\_\_\_\_ bags



## EXEMPLARY RESPONSE

38

Leia has 5 pounds of chocolate that she will put into bags. She puts  $\frac{1}{3}$  pound of the chocolate into each bag. Into how many bags does Leia put the chocolate?

*Answer*    15    bags

# GUIDE PAPER 1

38

Leia has 5 pounds of chocolate that she will put into bags. She puts  $\frac{1}{3}$  pound of the chocolate into each bag. Into how many bags does Leia put the chocolate?

$$5 \div \frac{1}{3} = 15$$

**Answer**

She puts chocolate into 15 bags.

bags

**Score Credit 1 (out of 1 credit)**

A correct answer is provided.

## GUIDE PAPER 2

38

Leia has 5 pounds of chocolate that she will put into bags. She puts  $\frac{1}{3}$  pound of the chocolate into each bag. Into how many bags does Leia put the chocolate?

*Answer*  bags

**Score Credit 1 (out of 1 credit)**

A correct answer is provided.

## GUIDE PAPER 3

38

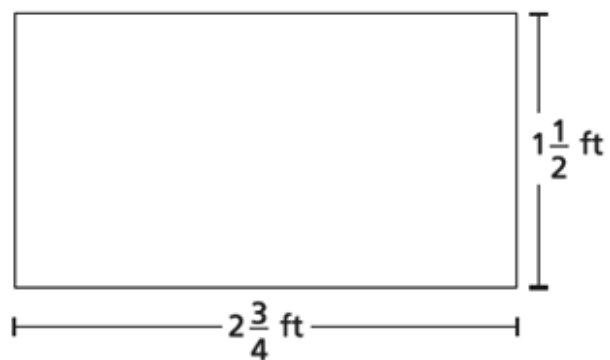
Leia has 5 pounds of chocolate that she will put into bags. She puts  $\frac{1}{3}$  pound of the chocolate into each bag. Into how many bags does Leia put the chocolate?

*Answer*  $5 \div \frac{1}{3} = \frac{1}{15}$  bags

**Score Credit 0 (out of 1 credit)**

An incorrect answer is provided.

A diagram of a rectangle with given dimensions is shown below.



What is the area, in square feet, of the rectangle?

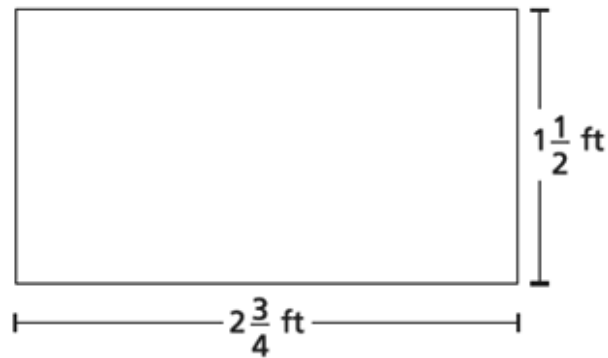
*Show your work.*

*Answer* \_\_\_\_\_ square feet

## EXEMPLARY RESPONSE

39

A diagram of a rectangle with given dimensions is shown below.



What is the area, in square feet, of the rectangle?

*Show your work.*

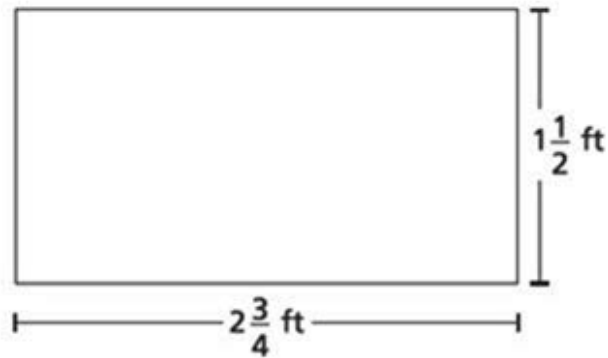
$$\begin{aligned} &1\frac{1}{2} \times 2\frac{3}{4} \\ &= \frac{3}{2} \times \frac{11}{4} \\ &= \frac{33}{8} \text{ OR } 4\frac{1}{8} \text{ square feet} \end{aligned}$$

*OR Other valid process*

$$\text{Answer } \frac{33}{8} \text{ OR } 4\frac{1}{8} \text{ square feet}$$

*OR Equivalent*

A diagram of a rectangle with given dimensions is shown below.



What is the area, in square feet, of the rectangle?

*Show your work.*

$$2\frac{3}{4} \times 1\frac{1}{2} =$$

$$\frac{11}{4} \times \frac{3}{2} = \frac{33}{8} = 4\frac{1}{8}$$

Answer

$$4\frac{1}{8}$$

square feet

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

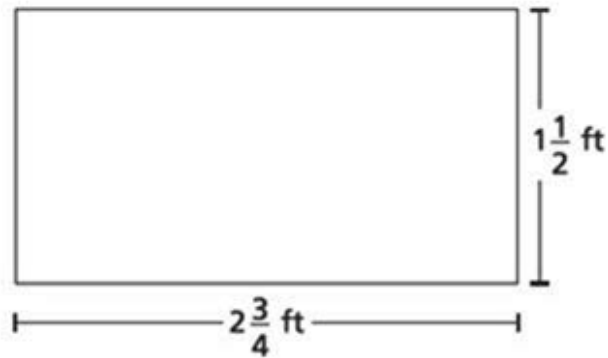
- The area of the rectangle is correctly determined using sound procedures.

This response is complete and correct.

## GUIDE PAPER 2

39

A diagram of a rectangle with given dimensions is shown below.



What is the area, in square feet, of the rectangle?

*Show your work.*

$4\frac{1}{8}$

Handwritten work showing the calculation of the area of the rectangle. The student uses a grid to multiply  $2\frac{3}{4}$  by  $1\frac{1}{2}$ . The grid shows the product as  $4\frac{1}{8}$ . To the right of the grid, the student shows the calculation:  $2\frac{3}{4} \times 1\frac{1}{2} = \frac{11}{4} \times \frac{3}{2} = \frac{33}{8} = 4\frac{1}{8}$ . The final answer is  $4\frac{1}{8}$ .

Answer  square feet

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The area of the rectangle is correctly determined using sound procedures.

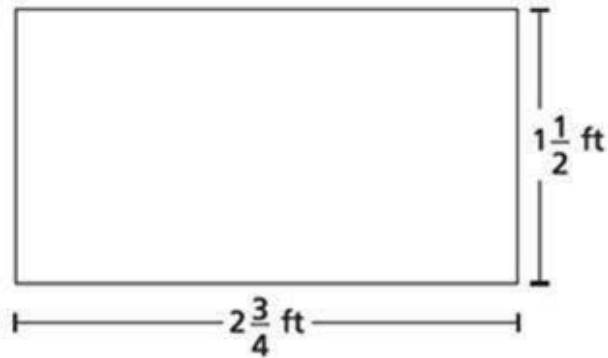
This response is complete and correct.



## GUIDE PAPER 3

39

A diagram of a rectangle with given dimensions is shown below.



What is the area, in square feet, of the rectangle?

*Show your work.*

$$\frac{11}{4} \times \frac{3}{2} = \frac{33}{8}$$

Answer

$$\frac{33}{8}$$

square feet

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

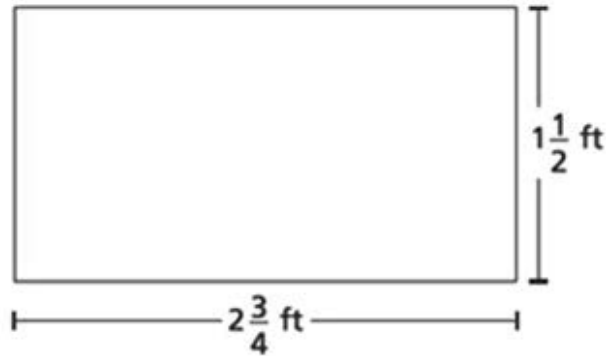
- The area of the rectangle is correctly determined using sound procedures.

This response contains sufficient work to demonstrate a thorough understanding.

## GUIDE PAPER 4

39

A diagram of a rectangle with given dimensions is shown below.



What is the area, in square feet, of the rectangle?

*Show your work.*

$$2\frac{3}{4} \times 1\frac{1}{2} = 4\frac{1}{8}$$

**Answer**

The area in square feet is  $4\frac{1}{8}$  . square feet

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

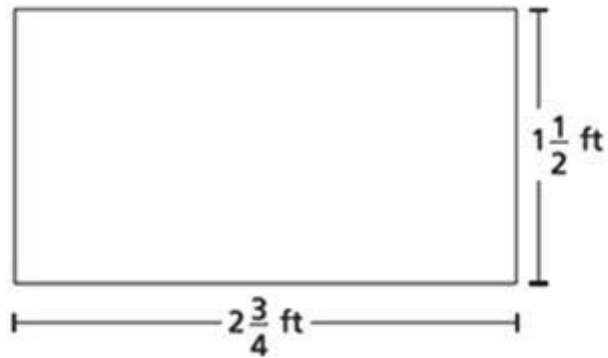
- Although a correct expression is written to determine the area of the rectangle and a correct solution is provided, it is not clear from the work how the solution is obtained.

This response contains the correct solution, but the required work is incomplete.

## GUIDE PAPER 5

39

A diagram of a rectangle with given dimensions is shown below.



What is the area, in square feet, of the rectangle?

*Show your work.*

$$\begin{array}{r}
 1\frac{1}{2} \rightarrow \frac{3}{2} \\
 \times 2\frac{3}{4} \rightarrow \frac{11}{4} \\
 \hline
 \frac{33}{8} = 4\frac{1}{8} \\
 \hline
 6\frac{1}{8}
 \end{array}$$

Answer  $6\frac{1}{4}$  square feet

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

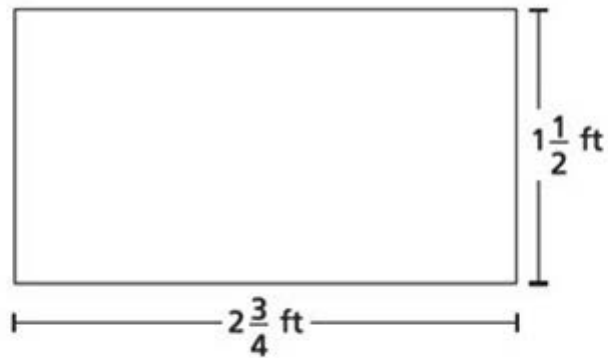
- The two dimensions are correctly written as improper fractions and correctly multiplied; however, errors are made when writing the solution as a mixed number.

This response contains an incorrect solution but applies a mathematically appropriate process.

## GUIDE PAPER 6

39

A diagram of a rectangle with given dimensions is shown below.



What is the area, in square feet, of the rectangle?

*Show your work.*

$$A = L \times W$$

$$A = 2\frac{3}{4} \times 1\frac{1}{2}$$

$$A = 2\frac{3}{8}$$

Answer  $2\frac{3}{8}$  square feet

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

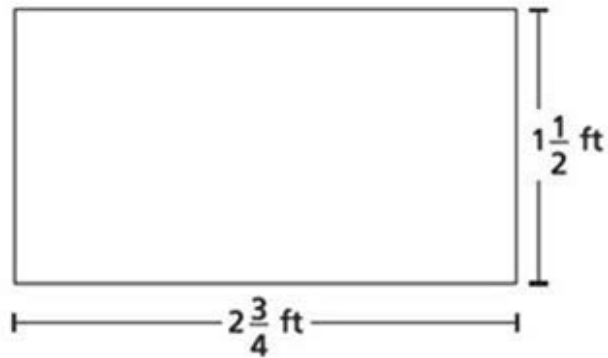
- A correct expression is written to determine the area of the rectangle; however, an incorrect solution is provided, and it is not clear how it is obtained.

This response correctly addresses only some elements of the task.

## GUIDE PAPER 7

39

A diagram of a rectangle with given dimensions is shown below.



What is the area, in square feet, of the rectangle?

*Show your work.*

$$4\frac{1}{8}$$

Answer  $4\frac{1}{8}$  square feet

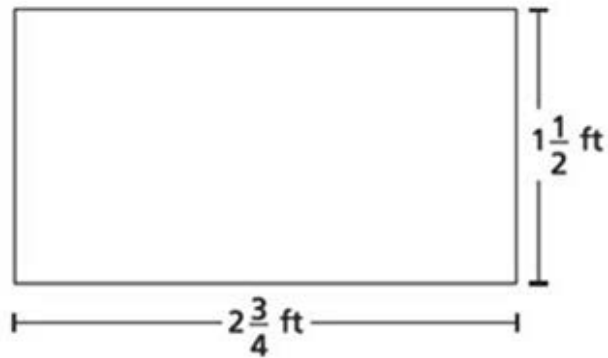
### Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- A correct solution is provided with no work to support it.

Per Scoring Policy #3 for 2- and 3-credit responses, this response receives no credit.

A diagram of a rectangle with given dimensions is shown below.



What is the area, in square feet, of the rectangle?

*Show your work.*

$$2\frac{3}{4} \times 2\frac{3}{4}$$

Answer  $5\frac{7}{10}$  square feet

### Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- An incorrect expression is written, and the numbers are incorrectly multiplied.

This response is incorrect and is insufficient to show any understanding.

Write a comparison statement using  $>$ ,  $<$ , or  $=$  that shows the relationship between the numbers 157.890 and 157.809.

*Explain how you know your answer is correct.*

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## EXEMPLARY RESPONSE

40

Write a comparison statement using  $>$ ,  $<$ , or  $=$  that shows the relationship between the numbers 157.890 and 157.809.

*Explain how you know your answer is correct.*

**157.890 > 157.809**

**The value of the digit in the hundredths place is larger in 157.890 than in 157.809.**

**I need to use the hundredths place to compare these numbers because they both have the same whole number and tenths value.**

***OR Other valid explanation***



Write a comparison statement using  $>$ ,  $<$ , or  $=$  that shows the relationship between the numbers 157.890 and 157.809.

*Explain how you know your answer is correct.*

157.890  $>$  157.809. I know I am correct because the digits in both the numbers are the same, up until the hundredths place. The digit in the hundredths place of 157.890 is nine , but the digit in the hundredths place of 157.809 is zero.  $9 > 0$ , so I know I am correct.

**Score Credit 2 (out of 2 credits)**

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- A correct comparison is written, and a correct explanation is provided comparing digits in the same place values in the two numbers.

The explanation is complete and correct.

## GUIDE PAPER 2

40

Write a comparison statement using  $>$ ,  $<$ , or  $=$  that shows the relationship between the numbers 157.890 and 157.809.

*Explain how you know your answer is correct.*

157.890  $>$  157.809 i know this because going from left to right all of the digits are the same all but the digit 9 the 9 in 157.890 is worth 0.090 and in 157.809 the 9 is worth 0.009 and 0.090 is ten times as much as 0.009

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- A correct comparison is written, and a correct explanation is provided comparing values of digit 9 in the two numbers.

The explanation is complete and correct.

## GUIDE PAPER 3

40

Write a comparison statement using  $>$ ,  $<$ , or  $=$  that shows the relationship between the numbers 157.890 and 157.809.

*Explain how you know your answer is correct.*

157.890  $>$  157.809      how i now is because in the hundredths place for the first one it has a nine and in the other hundredths it has a zero so nine is bigger than zero

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- A correct comparison is written, and a correct explanation is provided comparing digits in the hundredths place in the two numbers.

Although the place values of the first four digits in the two numbers are not addressed, the explanation is sufficient to show a thorough understanding.

## GUIDE PAPER 4

40

Write a comparison statement using  $>$ ,  $<$ , or  $=$  that shows the relationship between the numbers 157.890 and 157.809.

*Explain how you know your answer is correct.*

157.890  $>$  157.809 because the 9 in 157.809 is in thousandths but the 9 in 157.890 hundredths

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A correct comparison is written.
- However, the explanation does not sufficiently compare the value of digits in the hundredths place in the two numbers.

This response correctly addresses only some elements of the task.

## GUIDE PAPER 5

40

Write a comparison statement using  $>$ ,  $<$ , or  $=$  that shows the relationship between the numbers 157.890 and 157.809.

*Explain how you know your answer is correct.*

I got my answer by comparing the so I did  $157.890 > 157.809$  because 9 is greater than 0.

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A correct comparison is written.
- The digits 9 and 0 are compared; however, the place values of digits used in the comparison is unclear.

This response correctly addresses only some elements of the task.

## GUIDE PAPER 6

40

Write a comparison statement using  $>$ ,  $<$ , or  $=$  that shows the relationship between the numbers 157.890 and 157.809.

*Explain how you know your answer is correct.*

157.890  $>$  157.809

my answer is correct because the 0 in 157.890 is in the thousandths place where as the 0 in 157.809 is in the hundredths place.

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A correct comparison is written.
- However, the explanation does not sufficiently compare the value of digits in the hundredths place in the two numbers.

This response correctly addresses only some elements of the task.

## GUIDE PAPER 7

40

Write a comparison statement using  $>$ ,  $<$ , or  $=$  that shows the relationship between the numbers 157.890 and 157.809.

*Explain how you know your answer is correct.*

157.890  $>$  157.809

### Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- Although a correct comparison is written, no explanation is provided.

Per Scoring Policy #3 for 2- and 3-credit responses, this response receives no credit.

Write a comparison statement using  $>$ ,  $<$ , or  $=$  that shows the relationship between the numbers 157.890 and 157.809.

*Explain how you know your answer is correct.*

157.890 is correct because the nine's place is in the tens and in 157.809 the nine's place is in the one. So I know that 157.890 is correct.

**Score Credit 0 (out of 2 credits)**

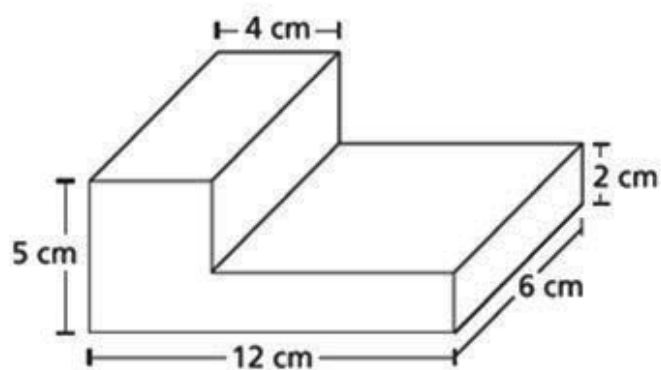
This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- A comparison is not written, and an incorrect explanation is provided.

The explanation is insufficient to show any understanding.



Two right rectangular prisms were combined to make the figure shown below.



What is the total volume, in cubic centimeters, of the figure?

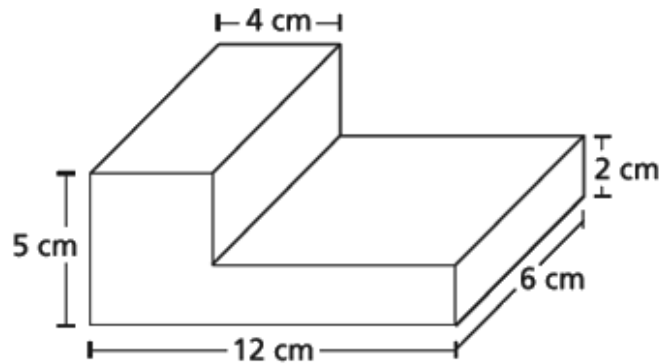
*Show your work.*

Answer \_\_\_\_\_ cubic centimeters

## EXEMPLARY RESPONSE

41

Two right rectangular prisms were combined to make the figure shown below.



What is the total volume, in cubic centimeters, of the figure?

*Show your work.*

$$5 \times 4 \times 6 = 120$$

$$8 \times 2 \times 6 = 96$$

$$120 + 96 = 216$$

*OR*

$$2 \times 12 \times 6 = 144$$

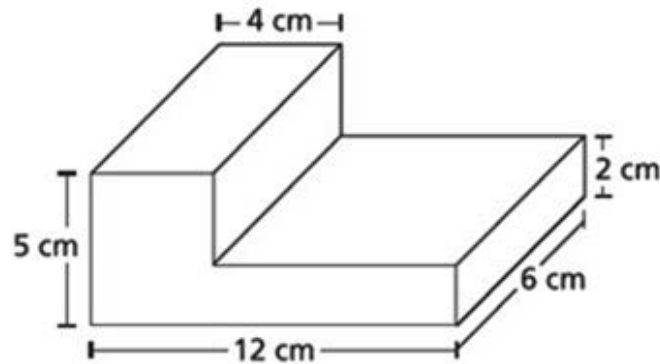
$$3 \times 4 \times 6 = 72$$

$$144 + 72 = 216$$

*OR Other valid process*

Answer 216 cubic centimeters

Two right rectangular prisms were combined to make the figure shown below.



What is the total volume, in cubic centimeters, of the figure?

*Show your work.*

$$V = l \times w \times h$$

$$V = 12 \times 6 \times 2$$

$$V = 144 \text{ cm}^3$$

$$V =$$

$$V = 3 \times 6 \times 4$$

$$V = 72 \text{ cm}^3$$

$$\begin{array}{r} 144 \\ + 72 \\ \hline 216 \text{ cm}^3 \end{array}$$

Answer  cubic centimeters

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

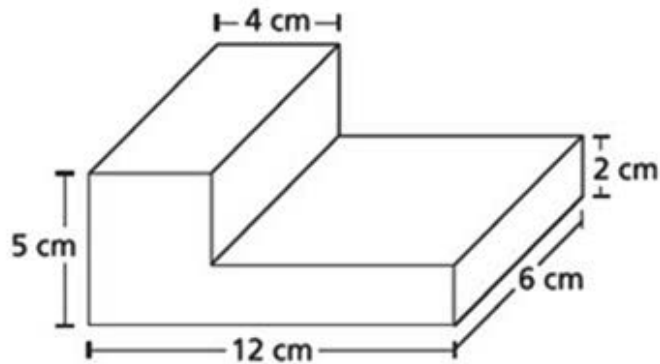
- The volume of each prism and the total volume are correctly determined using sound procedures.

This response is complete and correct.

## GUIDE PAPER 2

41

Two right rectangular prisms were combined to make the figure shown below.



What is the total volume, in cubic centimeters, of the figure?

*Show your work.*

$$\begin{array}{r} A = 4 \times 5 \times 6 = 120 \\ B = 8 \times 2 \times 6 = 96 \\ + \quad 120 \\ \quad 96 \\ \hline 216 \end{array}$$

Answer  cubic centimeters

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

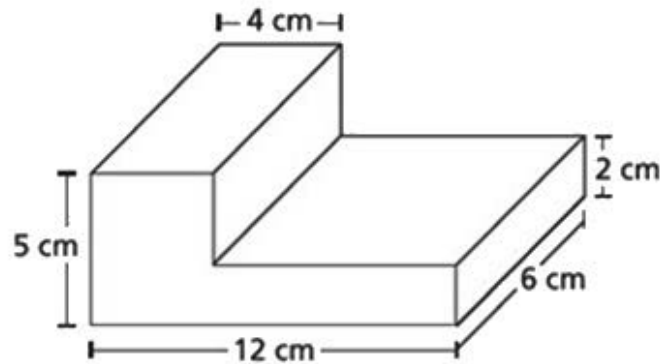
- The volume of each prism and the total volume are correctly determined using sound procedures.

This response is complete and correct.

## GUIDE PAPER 3

41

Two right rectangular prisms were combined to make the figure shown below.



What is the total volume, in cubic centimeters, of the figure?

*Show your work.*

|                           |                           |
|---------------------------|---------------------------|
| $V = L \times H \times W$ | $V = L \times H \times W$ |
| $V = 5 \times 4 \times 6$ | $V = 2 \times 6 \times 8$ |
| $V = 120$                 | $V = 96$                  |

Answer  cubic centimeters

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

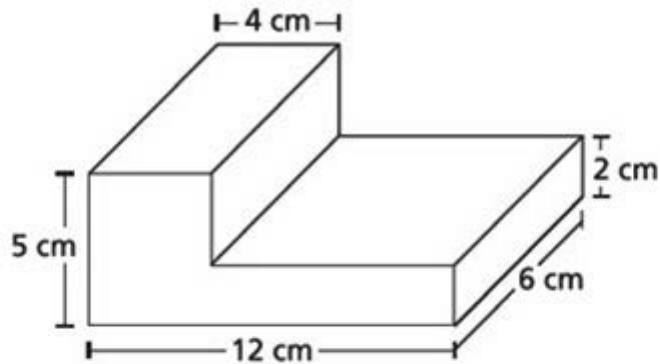
- The volume of each prism and the total volume are correctly determined using sound procedures.
- Although the last step of adding the two volumes is not shown, this does not detract from the demonstration of a thorough understanding.

This response contains sufficient work to show a thorough understanding.

## GUIDE PAPER 4

41

Two right rectangular prisms were combined to make the figure shown below.



What is the total volume, in cubic centimeters, of the figure?

*Show your work.*

$$A) 4 \times 5 \times 6 = 120$$

$$B) 6 \times 2 \times 6 = 72$$

$$120 + 72 = 192$$

Answer  cubic centimeters

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

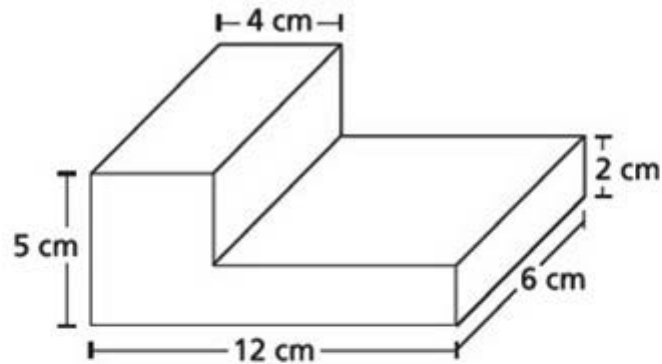
- The volume of the left part of the figure is correctly determined; however, an incorrect dimension of 6 instead of 8 is used when calculating the volume of the right part of the figure.

This response correctly addresses only some elements of the task.

## GUIDE PAPER 5

41

Two right rectangular prisms were combined to make the figure shown below.



What is the total volume, in cubic centimeters, of the figure?

*Show your work.*

$$(4 \times 6) \times 5 = 120$$

$$(8 \times 6) \times 2 = 76$$

$$\begin{array}{r} + 120 \\ 76 \\ \hline 196 \end{array}$$

Answer  cubic centimeters

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

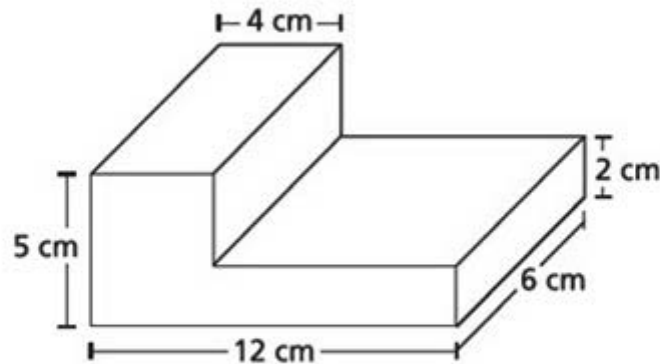
- A multiplication error occurs when calculating the volume of the right part of the figure, resulting in an incorrect solution.

This response contains an incorrect solution but applies a mathematically appropriate process.

## GUIDE PAPER 6

41

Two right rectangular prisms were combined to make the figure shown below.



What is the total volume, in cubic centimeters, of the figure?

*Show your work.*

( I cut the shape into 2 pieces and seperated the twelve so they're 4 & 8.)

$$8 \times 6 = 48, \quad 48 \times 2 = 96 \text{ cm.}$$

$$4 \times 5 = 20, \quad 20 \times 4 = 80 \text{ cm.}$$

$$96 + 80 = 176 \text{ cm.}$$

Answer  cubic centimeters

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The volume of the right part of the figure is correctly determined; however, an incorrect dimension of 4 instead of 6 is used when calculating the volume of the left part of the figure, resulting in an incorrect solution.
- The two calculated volumes are correctly added to determine the total volume.

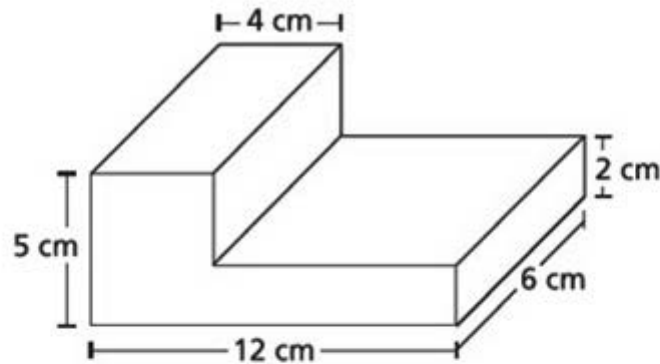
This response correctly addresses only some elements of the task.



## GUIDE PAPER 7

41

Two right rectangular prisms were combined to make the figure shown below.



What is the total volume, in cubic centimeters, of the figure?

*Show your work.*

$$\begin{aligned} 5 \times 4 \times 12 &= 240 \text{ cm} & 240 + 24 &= 264 \text{ cubic centimeters} \\ 12 \times 6 \times 2 &= 24 \text{ cm} \end{aligned}$$

**Answer** The total volume of the figure is 264 cubic centimeters. cubic centimeters

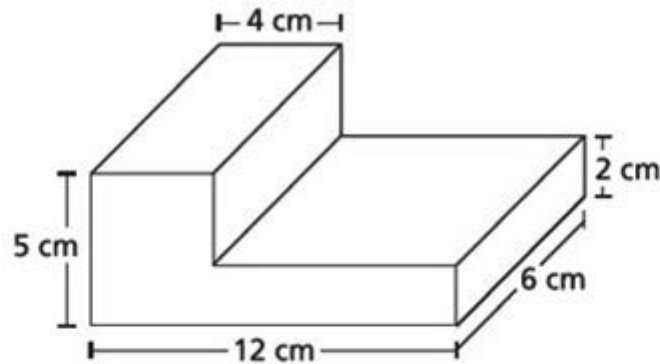
### Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- Although a correct expression is written to determine the volume of the bottom part of the figure, it is incorrectly calculated.
- An irrelevant volume of  $5 \times 4 \times 12$  figure is calculated and added to the other volume.

Holistically, this response is insufficient to show any understanding.

Two right rectangular prisms were combined to make the figure shown below.



What is the total volume, in cubic centimeters, of the figure?

*Show your work.*

$$12 \times 6 \times 2 = 144$$

Answer  cubic centimeters

### Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- Although the volume of the bottom part of the figure is correctly determined, it is inappropriately provided as a solution.
- The volume of the top of the figure and the total volume are not addressed.

Holistically, this response is insufficient to show any understanding.

A teacher writes the equation  $6 \times \frac{3}{3} = 6$  on the board. A student says the equation is wrong because multiplying 6 by a fraction results in a product that is less than 6. Is the student correct?

*Explain your answer.*

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## EXEMPLARY RESPONSE

42

A teacher writes the equation  $6 \times \frac{3}{3} = 6$  on the board. A student says the equation is wrong because multiplying 6 by a fraction results in a product that is less than 6. Is the student correct?

*Explain your answer.*

**The student is incorrect. When you multiply a number by a fraction equivalent to 1 (like  $\frac{3}{3}$ ) the number remains the same. It will not increase or decrease.**

***OR***

**No, the student is incorrect. The only time multiplying 6 by a fraction would result in a product less than 6 is when the fraction is less than 1. The fraction  $\frac{3}{3}$  is equal to 1, so  $6 \times \frac{3}{3}$  will be the same as  $6 \times 1$  which is 6. The teacher's equation is correct.**

***OR Other valid explanation***

42

A teacher writes the equation  $6 \times \frac{3}{3} = 6$  on the board. A student says the equation is wrong because multiplying 6 by a fraction results in a product that is less than 6. Is the student correct?

*Explain your answer.*

The student is incorrect. When multiplying 6 by a fraction less than one, that product will be less than six. When you multiply 6 by a fraction that is equal to one, then the answer will be 6.

**Score Credit 2 (out of 2 credits)**

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The student's claim is correctly identified, and multiplication by a fraction is correctly explained. The explanation is complete and correct.

## GUIDE PAPER 2

42

A teacher writes the equation  $6 \times \frac{3}{3} = 6$  on the board. A student says the equation is wrong because multiplying 6 by a fraction results in a product that is less than 6. Is the student correct?

*Explain your answer.*

no the student is incorrect because  $\frac{3}{3}$  is equal to one and  $1 \times 6 = 6$

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The student's claim is correctly identified, and multiplication by  $\frac{3}{3}$  is correctly explained. The explanation is complete and correct.

## GUIDE PAPER 3

42

A teacher writes the equation  $6 \times \frac{3}{3} = 6$  on the board. A student says the equation is wrong because multiplying 6 by a fraction results in a product that is less than 6. Is the student correct?

*Explain your answer.*

The Student is not correct because  $\frac{18}{3}$  is equal to 6.

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The student's claim is correctly identified, and multiplication by  $\frac{3}{3}$  is correctly explained. The explanation is sufficient to show a thorough understanding.

## GUIDE PAPER 4

42

A teacher writes the equation  $6 \times \frac{3}{3} = 6$  on the board. A student says the equation is wrong because multiplying 6 by a fraction results in a product that is less than 6. Is the student correct?

*Explain your answer.*

The student is not correct because  $\frac{3}{3} = 1$  so  $6 \times \frac{3}{3} = 6$ .

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The student's claim is correctly identified, and  $\frac{3}{3}$  is correctly compared with 1; however, an incorrect product is written when explaining the multiplication.

This response correctly addresses only some elements of the task.



## GUIDE PAPER 5

42

A teacher writes the equation  $6 \times \frac{3}{3} = 6$  on the board. A student says the equation is wrong because multiplying 6 by a fraction results in a product that is less than 6. Is the student correct?

*Explain your answer.*

$$\frac{3}{3} = 1 \quad 1 \times 6 = 6$$

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The multiplication by  $\frac{3}{3}$  is correctly explained; however, the student's claim is not identified as incorrect.

This response correctly addresses only some elements of the task.

## GUIDE PAPER 6

42

A teacher writes the equation  $6 \times \frac{3}{3} = 6$  on the board. A student says the equation is wrong because multiplying 6 by a fraction results in a product that is less than 6. Is the student correct?

*Explain your answer.*

no because  $\frac{3}{3}$  is = 1

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The student's claim is correctly identified, and  $\frac{3}{3}$  is correctly compared with 1; however, multiplication by  $\frac{3}{3}$  is not clearly explained.

This response correctly addresses only some elements of the task.

## GUIDE PAPER 7

42

A teacher writes the equation  $6 \times \frac{3}{3} = 6$  on the board. A student says the equation is wrong because multiplying 6 by a fraction results in a product that is less than 6. Is the student correct?

*Explain your answer.*

the answer is 7 because you have to multiply  $\frac{6}{1} \times \frac{3}{3} =$   
so you have to do  $3 \times 1 = 3$  so what is  $3 \times 6$  its 18 and  
then  $3 \times 3$  so its 9 and its  $\frac{9}{3} + \frac{18}{3} = \frac{27}{3}$   
and then you have to  $27 \div 3 = 7$  the answer is 7

### Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- The student's claim is not identified as correct or incorrect, and an incorrect explanation is provided. The explanation is insufficient to show any understanding.

42

A teacher writes the equation  $6 \times \frac{3}{3} = 6$  on the board. A student says the equation is wrong because multiplying 6 by a fraction results in a product that is less than 6. Is the student correct?

*Explain your answer.*

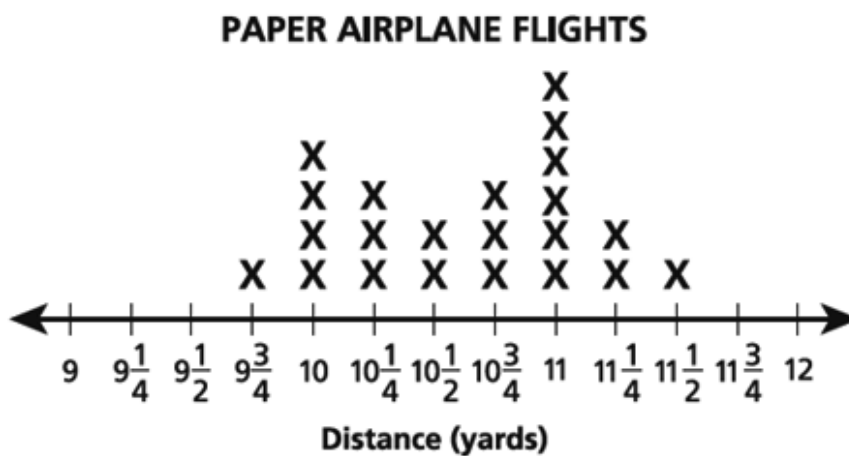
yes i think the student is corrt because mutiplying 6 by a faction results in a product that is less then 6.

**Score Credit 0 (out of 2 credits)**

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- The student's claim is incorrectly identified, and multiplication by a fraction is not explained. The explanation is insufficient to show any understanding.

The Science Club tests paper airplane designs by measuring how far they fly. The results are recorded in the line plot shown below.



What is the difference, in yards, between the longest and shortest flights?

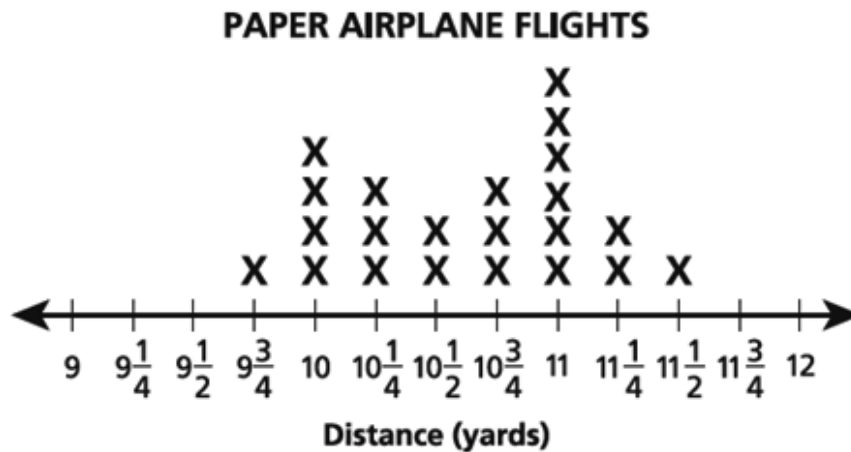
*Show your work.*

**Answer** \_\_\_\_\_ yards

# EXEMPLARY RESPONSE

43

The Science Club tests paper airplane designs by measuring how far they fly. The results are recorded in the line plot shown below.



What is the difference, in yards, between the longest and shortest flights?

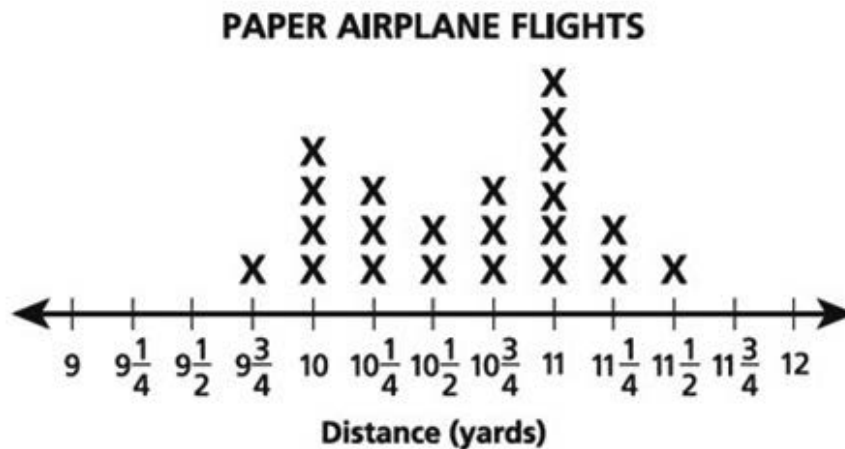
*Show your work.*

$$11\frac{1}{2} - 9\frac{3}{4} = 11\frac{2}{4} - 9\frac{3}{4} = 10\frac{6}{4} - 9\frac{3}{4} = 1\frac{3}{4}$$

*OR Other valid process*

Answer  $\underline{1\frac{3}{4} \text{ OR } \frac{7}{4}}$  yards  
*OR Equivalent*

The Science Club tests paper airplane designs by measuring how far they fly. The results are recorded in the line plot shown below.



What is the difference, in yards, between the longest and shortest flights?

*Show your work.*

$$\begin{array}{r}
 11\frac{2}{4} = 10\frac{6}{4} \\
 - \quad \quad 9\frac{3}{4} \\
 \hline
 1\frac{3}{4}
 \end{array}$$

Answer  $1\frac{3}{4}$  yards

## Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

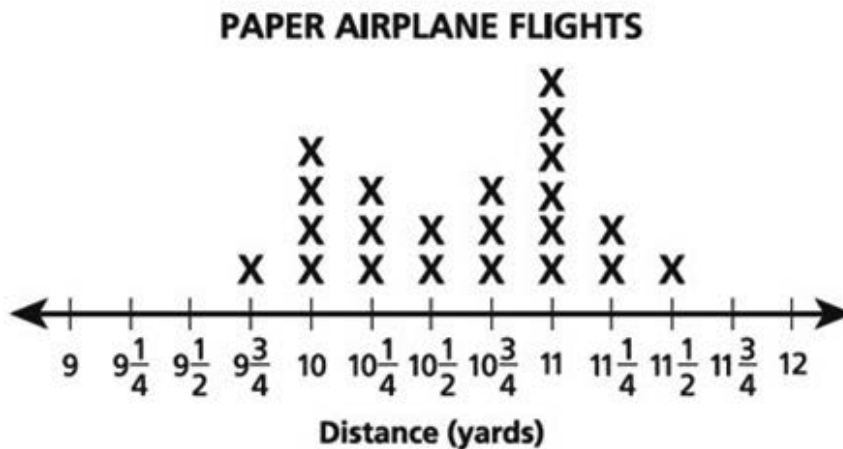
- The longest and shortest flights are correctly identified, and the difference between them is correctly determined using sound procedures.

This response is complete and correct.

## GUIDE PAPER 2

43

The Science Club tests paper airplane designs by measuring how far they fly. The results are recorded in the line plot shown below.



What is the difference, in yards, between the longest and shortest flights?

*Show your work.*

$$11\frac{1}{2} = \frac{23}{2} \text{ and } 9\frac{3}{4} = \frac{39}{4}, \frac{23}{2} = \frac{46}{4} \quad 46 - 39 = 7,$$

the difference is  $\frac{7}{4}$ , or  $1\frac{3}{4}$ .

Answer  $1\frac{3}{4}$  yards

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The longest and shortest flights are correctly identified, and the difference between them is correctly determined using sound procedures.

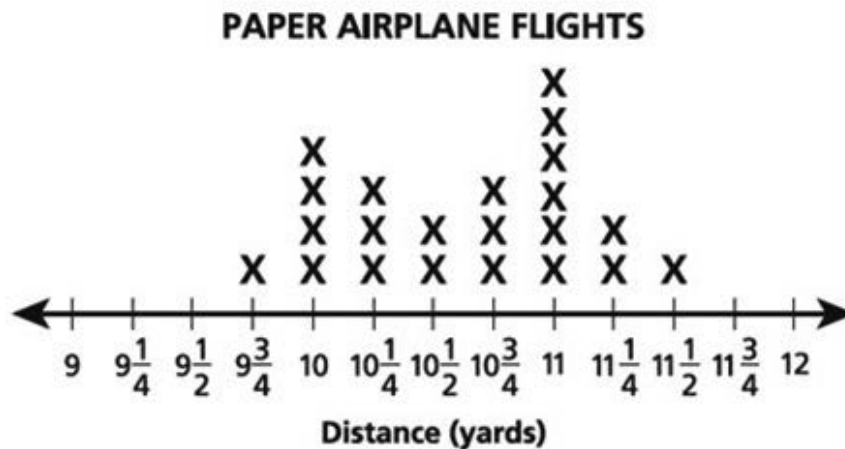
This response is complete and correct.



## GUIDE PAPER 3

43

The Science Club tests paper airplane designs by measuring how far they fly. The results are recorded in the line plot shown below.



What is the difference, in yards, between the longest and shortest flights?

*Show your work.*

$$\begin{array}{rcl} \text{Longest} & = & 11\frac{1}{2} = 11\frac{2}{4} \\ \text{Shortest} & = & 9\frac{3}{4} = \quad - \quad 9\frac{3}{4} \\ & & \text{-----} \\ & & 1\frac{3}{4} \end{array}$$

Answer  $1\frac{3}{4}$  yards

### Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

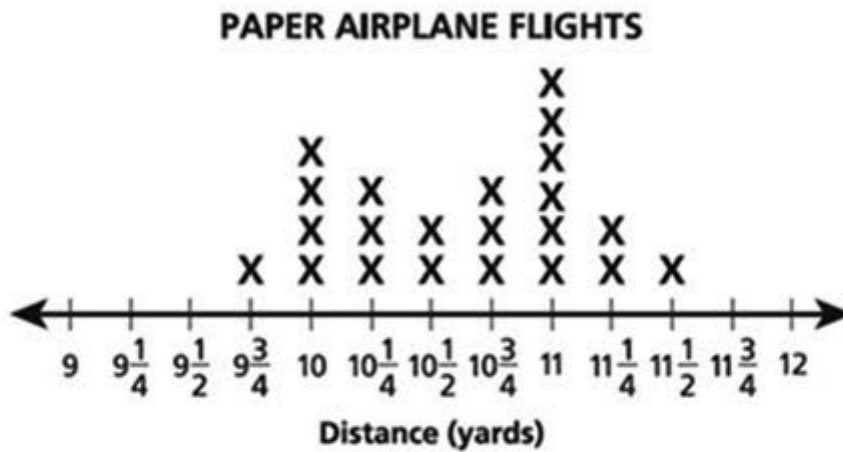
- The longest and shortest flights are correctly identified, and the difference between them is correctly determined using sound procedures.

This response contains sufficient work to demonstrate a thorough understanding.

## GUIDE PAPER 4

43

The Science Club tests paper airplane designs by measuring how far they fly. The results are recorded in the line plot shown below.



What is the difference, in yards, between the longest and shortest flights?

*Show your work.*

$$11\frac{1}{2} - 9\frac{3}{4} =$$

Answer  yards

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

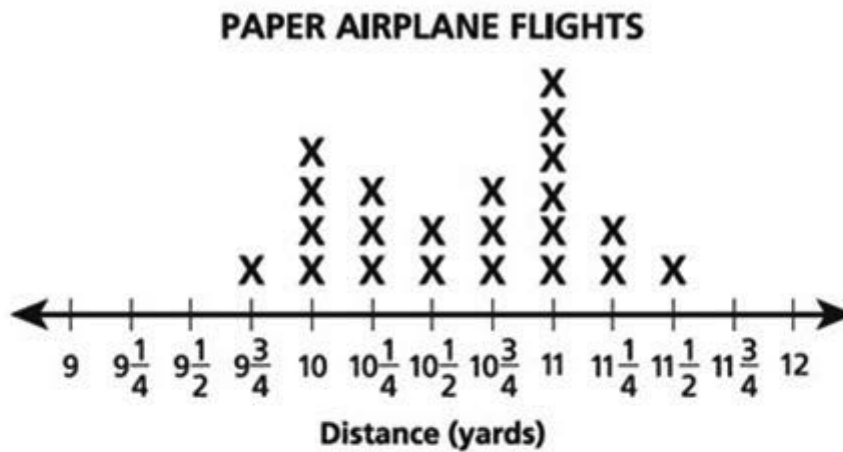
- The longest and shortest flights are correctly identified, and a correct expression is written to determine the difference; however, the difference is not calculated.

This response correctly addresses only some elements of the task.

## GUIDE PAPER 5

43

The Science Club tests paper airplane designs by measuring how far they fly. The results are recorded in the line plot shown below.



What is the difference, in yards, between the longest and shortest flights?

*Show your work.*

$$11 - 9\frac{3}{4} = 1\frac{1}{4}$$

Answer  $1\frac{1}{4}$  yards

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

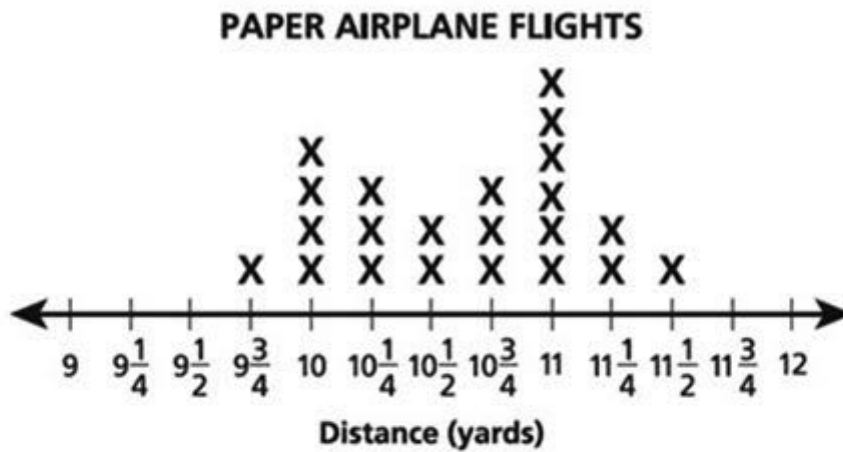
- The shortest flight is correctly identified; however, the longest flight is incorrect.
- The two distances are correctly subtracted to determine the difference.

This response correctly addresses only some elements of the task.

## GUIDE PAPER 6

43

The Science Club tests paper airplane designs by measuring how far they fly. The results are recorded in the line plot shown below.



What is the difference, in yards, between the longest and shortest flights?

*Show your work.*

$$9\frac{3}{4} - 11\frac{1}{2}$$

Answer  $1\frac{3}{4}$  yards

### Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

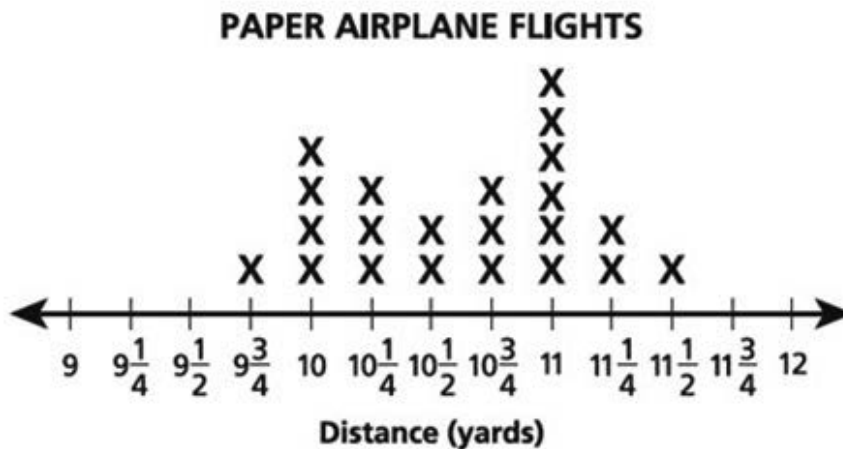
- The longest and shortest flights are correctly identified, and a correct solution is provided; however, the subtraction is written in an incorrect order.

This response correctly addresses only some elements of the task.

# GUIDE PAPER 7

43

The Science Club tests paper airplane designs by measuring how far they fly. The results are recorded in the line plot shown below.



What is the difference, in yards, between the longest and shortest flights?

*Show your work.*

$$9\frac{3}{4} - 11\frac{1}{2}$$

$$\frac{6}{2} - \frac{7}{2} = \frac{1}{2}$$

$$9 - 9 = 0$$

Answer  $\frac{1}{2}$  yards

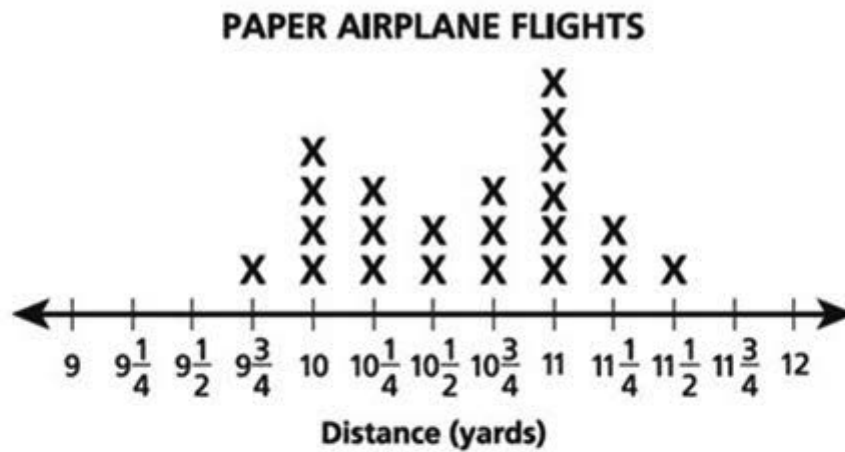
## Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- Although the longest and shortest flights are correctly identified, the subtraction is written in an incorrect order and is incorrectly carried out.

Holistically, this response is insufficient to show any understanding.

The Science Club tests paper airplane designs by measuring how far they fly. The results are recorded in the line plot shown below.



What is the difference, in yards, between the longest and shortest flights?

*Show your work.*

$$9\frac{3}{4} \div 11 = 1\frac{3}{4}$$

Answer the difference is  $1\frac{3}{4}$  yards yards

### Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- Although the correct solution is provided, it is obtained using an obviously incorrect procedure. Holistically, this response is insufficient to show any understanding.

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

**Answer** \_\_\_\_\_ miles

## EXEMPLARY RESPONSE

44

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

**Josh ran 12.35 miles in March. To find the number of miles he ran in April, multiply the number of miles he ran in March, 12.35 times 3 for a total of 37.05 miles in April. To find the number of miles he ran in May, add 43.1 to the number of miles he ran in March, 12.35, for a total of 55.45 miles. To find the total number of miles, add the distances for these three months together,  $12.35 + 37.05 + 55.45$ , for a total of 104.85 miles.**

***OR***

**March: 12.35**

**April:  $12.35 \times 3 = 37.05$**

**May:  $43.1 + 12.35 = 55.45$**

**Total:  $12.35 + 37.05 + 55.45 = 104.85$**

***OR Other valid process***

**Answer 104.85 miles**



Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

The student's work is shown in a box. On the left, there is a multiplication problem:  $12.35 \times 3 = 37.05$ . Below this, the student lists: 12.35 - March, 37.05 - April, and 55.45 - May. On the right, there are two addition problems. The first is  $12.35 + 43.10 = 55.45$ . The second is a sum of three values:  $12.35 + 37.05 + 55.45 = 104.85$ .

Answer

104.85 miles

### Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The number of miles Josh runs in April and May is correctly calculated, and the total number of miles he runs for three months is correctly determined using sound procedures.

This response is complete and correct.

## GUIDE PAPER 2

44

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

|                                                                  |                                                                 |                                                                 |                                                                  |
|------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------|
| $\begin{array}{r} \times 12.35 \\ 3 \\ \hline 37.05 \end{array}$ | $\begin{array}{r} + 12.35 \\ 43.10 \\ \hline 55.45 \end{array}$ | $\begin{array}{r} + 12.35 \\ 37.05 \\ \hline 49.40 \end{array}$ | $\begin{array}{r} + 55.45 \\ 49.40 \\ \hline 104.85 \end{array}$ |
|------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------|

**Answer** Josh ran 104.85 miles. miles

### Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The number of miles Josh runs in April and May is correctly calculated, and the total number of miles he runs for three months is correctly determined using sound procedures.

This response is complete and correct.

## GUIDE PAPER 3

44

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

$$(3 \times 12.35) + (43.1 + 12.35) + 12.35 = 104.85$$

*Answer*  miles

### Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The total number of miles Josh runs for three months is correctly determined using sound procedures. This response contains sufficient work to demonstrate a thorough understanding.

## GUIDE PAPER 4

44

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

The student's work is shown in three separate boxes. The first box shows the calculation for April:  $12.35 \times 100$  (crossed out) and  $12.35 \times 3$  resulting in  $37.05$ . The second box shows the calculation for May:  $12.35 + 43.10$  resulting in  $55.45$ . The third box shows the final sum:  $12.35 + 37.05 + 55.45 = 104.85$ .

Answer  miles

### Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- The number of miles Josh runs in April and May is correctly calculated; however, an error when determining the total number of miles results in an incorrect solution.

This response contains an incorrect solution but provides sound procedures.

## GUIDE PAPER 5

44

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

first,  $12.35 \times 3 = 37.05$

then,  $43.1 + 12.35 = 55.45$

finally,  $55.45 + 37.05 = 92.50$

Answer  miles

### Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- The number of miles Josh runs in April and May is correctly calculated; however, March miles are ignored when determining the total number of miles.

This response reflects some minor misunderstanding of the underlying mathematical concepts and procedures.

## GUIDE PAPER 6

44

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

|                                                                  |                                                                       |                                                                 |                                                                      |
|------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------|
| $\begin{array}{r} \times 12.35 \\ 3 \\ \hline 37.05 \end{array}$ | $\begin{array}{r} \times 12.35 \\ 43.1 \\ \hline 532.285 \end{array}$ | $\begin{array}{r} + 12.35 \\ 37.05 \\ \hline 49.40 \end{array}$ | $\begin{array}{r} + 532.285 \\ 49.400 \\ \hline 581.685 \end{array}$ |
|------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------|

Answer  miles

### Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- The number of miles Josh runs in April is correctly calculated; however, multiplication instead of addition is used when determining miles for May.
- The calculated miles for each month are correctly added to determine the total number of miles.

This response reflects some minor misunderstanding of the underlying mathematical concepts and procedures.

## GUIDE PAPER 7

44

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

$$\begin{aligned}12.35 \times 3 &= 37.05 \\12.35 + 43.10 &= 56.45 \\37.05 + 56.45 &= 93.50\end{aligned}$$

Answer  miles

### Score Credit 1 (out of 3 credits)

This response demonstrates only a limited understanding of the mathematical concepts and procedures in the task.

- The number of miles Josh runs in April is correctly calculated; however, an addition error occurs when calculating miles for May.
- March miles are ignored when determining the total number of miles.

This response addresses some elements of the task correctly but exhibits multiple flaws related to misunderstanding of important aspects of the task.

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

$$\begin{aligned}12.35 \times 3 &= 37.05 = \text{April} \\37.05 + 43.1 &= 80.15 = \text{May} \\12.35 + 37.05 + 80.15 &= 139.55 = \text{total}\end{aligned}$$

Answer  miles

### Score Credit 1 (out of 3 credits)

This response demonstrates only a limited understanding of the mathematical concepts and procedures in the task.

- The number of miles Josh runs in April is correctly calculated; however, April instead of March miles are used to determine miles for May.
- A calculation error is made when determining the total number of miles.

This response addresses some elements of the task correctly but exhibits multiple flaws related to misunderstanding of important aspects of the task.



## GUIDE PAPER 9

44

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

Show your work.

$$\begin{array}{r}
 12.35 \\
 + 12.35 \\
 + 12.35 \\
 \hline
 37.05
 \end{array}
 \qquad
 \begin{array}{r}
 37.05 \\
 + 43.10 \\
 \hline
 80.15
 \end{array}$$

Answer

80.15 miles in  
total

miles

### Score Credit 1 (out of 3 credits)

This response demonstrates only a limited understanding of the mathematical concepts and procedures in the task.

- The number of miles Josh runs in April is correctly calculated.
- March miles are ignored and 43.1 is inappropriately used for May miles when determining the total number of miles.

This response addresses some elements of the task correctly but provides reasoning that is faulty or incomplete.

## GUIDE PAPER 10

44

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

$$\begin{array}{r} 43.10 \\ 12.35 \\ + 03.00 \\ \hline 58.45 \end{array}$$

**Answer** Josh ran 58.45 miles in total. miles

### Score Credit 0 (out of 3 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- An incorrect procedure is used to obtain an incorrect solution.

This response is insufficient to show any understanding.

44

Josh is training for a race. The number of miles he runs each month for three months is shown below.

- Josh runs 12.35 miles in March.
- Josh runs 3 times as many miles in April as in March.
- Josh runs 43.1 more miles in May than he ran in March.

What is the total number of miles Josh runs for these three months?

*Show your work.*

Answer  miles

**Score Credit 0 (out of 3 credits)**

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- The correct solution is provided with no work.

Per Scoring Policy #3 for 2- and 3-credit responses, this response receives no credit.